CS4051
Information Retrieval
Week 01

Muhammad Rafi
January 19, 2026

Agenda

- Course Introduction
- Some Basic Terminology
- Difference between Database & IR
- Ad hoc IR setup
- IR Dimension
- Large Language Models
- Retrieval Argument Generations
- Conclusion

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Course Objectives

- Information retrieval is the activity of obtaining information resources relevant to an information need from a collection of information resources usually document collections.
- The course discusses important retrieval models (Boolean, vector space, probabilistic, inference net, language modeling, link analysis), clustering algorithms, collaborative filtering, automatic text categorization, and experimental evaluation.
- Search Engines Algorithms, Advanced IR techniques and Evaluations are discussed.

Course Outline (1/4)

- Week -01
  - Introduction to IR course, IR Problem and its Components, Basic IR model – Boolean Information Retrieval, Extended Boolean Model, Example of Commercial Systems – WestLaw <Ch: 1 >
- Week-02
  - Document Processing, Term vocabulary and positing lists, Stemming & Lemmatization, posting list processing via skip lists, Phrase Query, positional indexing, bi-word indexing, Combining different indexing techniques. <Ch: 2>

Course Outline (1/4)

- Week -03
  - Dictionary and Tolerant Retrieval, Search Structures for Dictionary, Wildcard queries, permuterm index, k-gram index, Spelling Correction, Edit Distance, Phonetic Correction <Ch: 3>
- Week-04
  - Index construction, single pass, in-memory, distributed indexing, dynamic indexing.
  - Heaps law, Zipf's law, dictionary indexing, fixed length and variable length coding, γ codes <Ch:4 and 5>

Course Outline (2/4)

- Week -05
  - Vector Space Model <Ch: 6 and Ch:7>
- Week-06
  - Midterm Exam
- Week-07
  - Evaluation of IR <Ch: 8>
- Week-08
  - Relevance Feedback <Ch: 9>

Course Outline (3/4)

- Week-9
  - Probabilistic Language Model for IR
  - Language Model for IR
- Week-10
  - Basic Web Search, Crawler and Indexing
  - Link Analysis <Ch: 19,20 and 21>
- Week-11
  - Midterm II
- Week-12
  - Text Classification/Clustering <Ch: 14,15, 16,17>

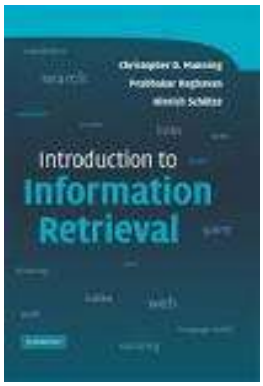
Course Outline (4/4)

- Week-13
  - Neural Information Retrieval
- Week-14
  - Learning Word Representation
- Week-15
  - RAG
- Week-16
  - Contrastive Learning

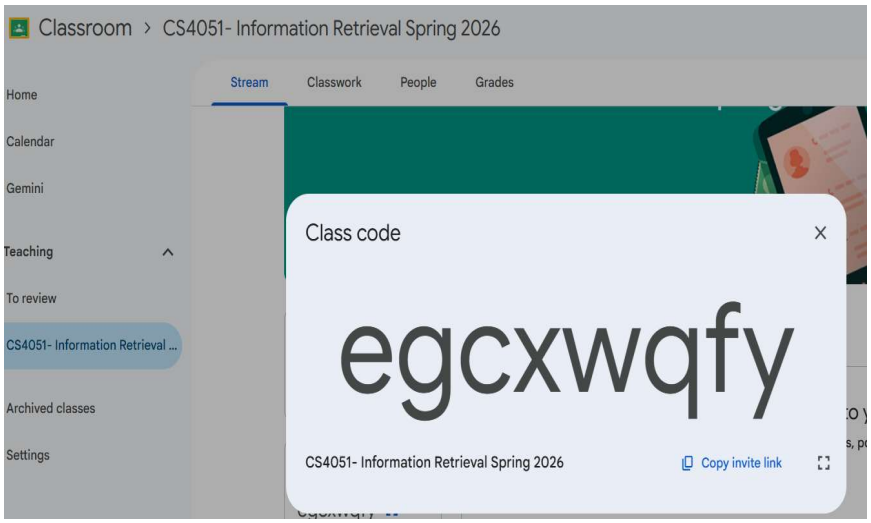
Grading Scheme

- Assignments 20%
- Quizzes 10%
- Mid-Terms (2) 20%
- Class Project 10%
- Final 40%

Text Books/References



GCR – Code



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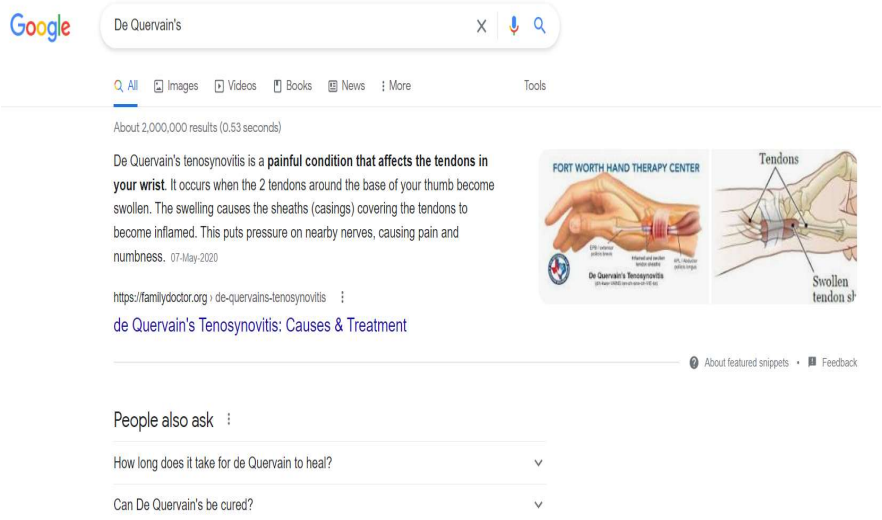
Basic Terminology

- Information Retrieval
- Data / Information / Knowledge/ Wisdom
- Database systems Vs. Information retrieval
- Data Mining Vs. Text mining
- Text Mining Vs. Information retrieval
- Ad hoc Information Retrieval

Information Retrieval

- Information Retrieval (IR) is activity of finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need (query) from within large digital collections (usually stored on computers or web).
- Example
  - What is your CNIC Number? CC?
  - What is De Quervain's?
  - How to get rid from kitchen bugs?
  - What is Surakav?
  - What is Sarcopenia?

Example

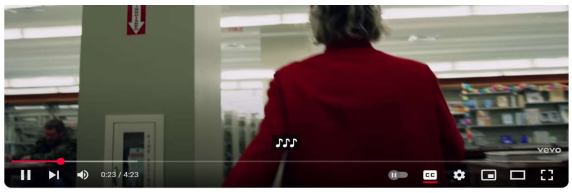


Data / Information / Knowledge/ Wisdom

- Data
  - Raw representation, symbols/signs
- Information
  - Processed data that is useful to answer questions(meaning of data in context) – Linked data
- Knowledge
  - Organized information that can be acted
- Wisdom
  - Applied knowledge DIKW Pyramid

DIKW Pyramid

Information Types



Leblanc_Leon.mp3

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Database vs. Retrieval

	Database	Information Retrieval
Retrieval	Structured data	Mostly unstructured
Query	SQL -Unambiguous	NLP - Imprecise
Result	Exact and Correct	Relevant or related
Interaction	Single Query Analytical Query	Interactive

Data Mining vs. Text Mining

Aspect	Text Mining	Data Mining
Definition	The process of extracting meaningful information from unstructured or semi-structured text data.	The process of discovering patterns, trends, and knowledge from structured data stored in databases or data warehouses.
Input Data	Unstructured or semi-structured text data (e.g., emails, reports, articles).	Structured numerical or categorical data (e.g., database tables, spreadsheets).
Techniques Used	Natural Language Processing (NLP), information retrieval, and linguistic analysis.	Statistical methods, machine learning, and database query techniques.
Challenges	Handling ambiguities, context understanding, and natural language variations.	Managing large datasets, ensuring data quality, and computational scalability.
Data Preprocessing	Involves tokenization, stemming, lemmatization, and stop-word removal.	Involves data cleaning, normalization, and transformation.

Information Retrieval vs. Text Mining

Aspect	Text Mining	Information Retrieval
Definition	The process of extracting meaningful insights and patterns from unstructured or semi-structured text data.	The process of retrieving relevant documents or information from a large repository in response to a query.
Input Data	Unstructured or semi-structured text data (e.g., documents, emails, articles).	Structured or unstructured data repositories, such as databases, web pages, or document collections.
Objective	Derive hidden patterns, insights, or relationships from text.	Find and rank the most relevant documents for a given query.
Techniques Used	Natural Language Processing (NLP), machine learning, and statistical analysis.	Search algorithms, indexing, ranking, and relevance feedback.
Data Preprocessing	Tokenization, stemming, lemmatization, and stop-word removal.	Index creation, keyword extraction, and document ranking.
Challenges	Handling language ambiguity, context, and polysemy.	Balancing precision and recall, handling diverse queries, and scalability.

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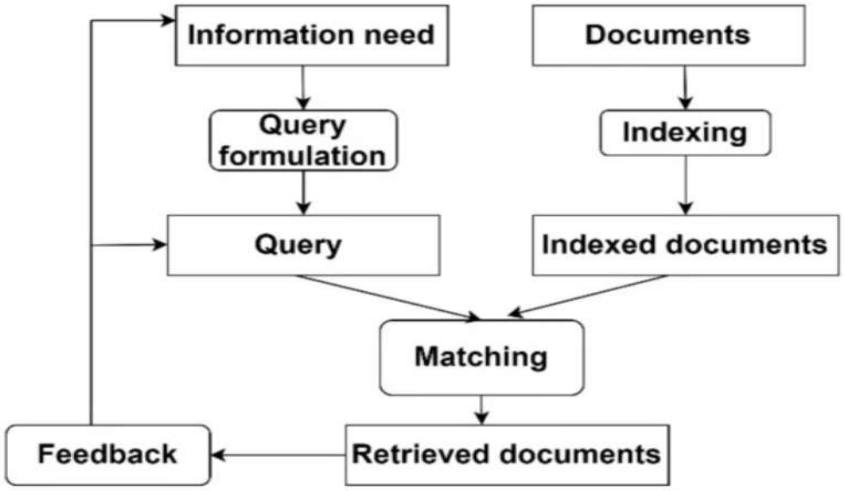
Adhoc Retrieval Systems

- It is a system that aims to provide documents from within the collection that are relevant to an arbitrary user information need, communicated to the system by means of a one-off, user-initiated query.
- Example
  - Google
  - Bing (later Decision engine)

Basic Information Retrieval Process



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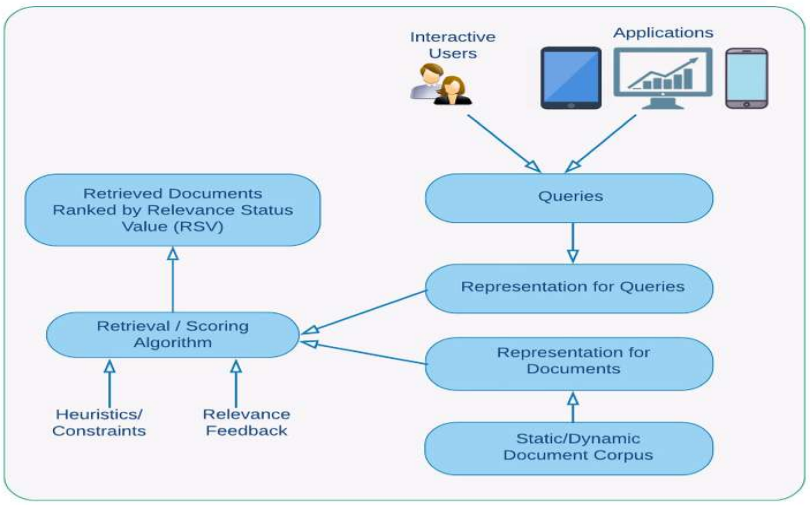

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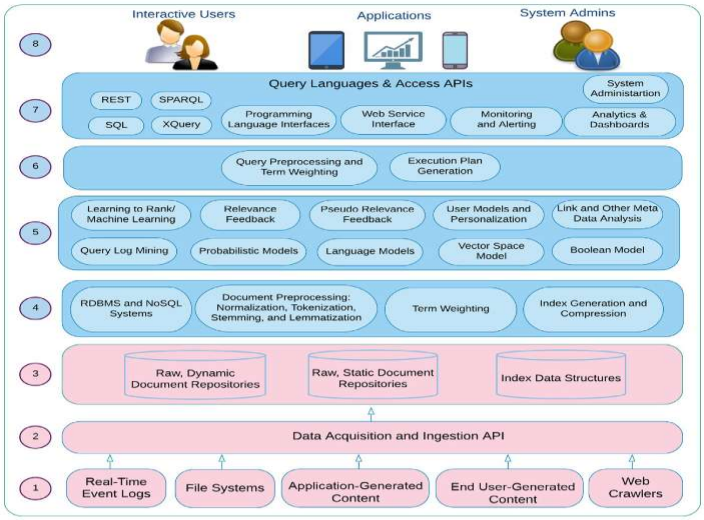

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Content wise IR Systems

- Contents
  - Size
    - Personal / Desktop (Spotlight, Instant Search,anywhere)
    - Web Scale (Google, DuckDuckgo)
    - Enterprise Search (WestLaw,PolicyBazar)
  - Static (Offline) or Dynamic (Online)
  - Type
    - Text
    - Multimedia
    - Mixed
  - Exact Match vs. Best Match

Dimension of IR

Content	Applications	Tasks
Text	Web search	Ad hoc search
Images	Vertical search	Filtering
Video	Enterprise search	Classification
Scanned docs	Desktop search	Question answering
Audio	Forum search	Clustering
Music	P2P search	
	Literature search	

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Large Language Model

- A Large Language Model (LLM) is a type of advanced AI model, typically based on deep learning architectures like transformers, that is trained on vast amounts of text data to understand and generate human-like text.
- It is a generative model which can generate a textual response against a prompt.
  - How to start ?
  - How to stop?
  - Most probable sequence of text ?

Large Language Model

Aspect	Large Language Model (LLM)	Information Retrieval (IR)
Functionality	Generates text, interprets queries, and performs reasoning.	Retrieves pre-existing documents or data based on relevance.
Generative Ability	Can create new text (e.g., generate stories or code).	Cannot generate; works only with pre-existing data.
Contextual Understanding	Handles complex linguistic context and reasoning.	Primarily focuses on matching queries with relevant documents.
Applications	Chatbots, text generation, summarization, etc.	Search engines, recommendation systems, etc.
Underlying Models	Based on deep learning (e.g., transformers).	Often uses traditional methods like TF-IDF, BM25, or embeddings.
Dynamic vs Static	Dynamic, as it generates content in real-time.	Static, operating over a fixed dataset.

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Retrieval-Augmented Generation (RAG)

- RAG bridges the gap between retrieval-based systems (IR) and generative models (LLMs)
- RAG is a framework that combines retrieval from an external knowledge source with a generative language model to improve the quality of generated responses.
- RAG enhances the generative model by grounding its responses in relevant retrieved documents.

How RAG Works?

- A query is input into the retriever.
- The retriever searches the knowledge base and retrieves the top-k relevant documents.
- These documents, along with the original query, are passed to the generator.
- The generator uses the retrieved context to produce a grounded response.

RAG vs. IR

- The main difference between RAG and IR lies in their objectives, methods, and outputs.

Feature	RAG	IR
Focus	Generating responses grounded in retrieved knowledge.	Retrieving and ranking relevant documents.
Components	Combines retrieval and generation.	Only retrieval.
Output	Human-like text response.	Pre-existing documents or snippets.
Best Use Case	Complex, conversational, or open-ended queries requiring reasoning.	Straightforward search for existing information.

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Music Search Engine

- Search by metadata likes: artists biography, music reviews, new releases, concerts dates
- Search for Lyrics: Lyrics.com. Allmusic.com musicpedia.com and SearchLyrics.com
- Recommend Similar Music: based on seed elements (artist, track) users are recommended similar tracks or artists.
- User query by Humming
- Generate Playlists: automatic generation of playlists, that satisfy user constraints

Mathematical Search Engines

- The mathematical contents on the web is continuously on rise. The IR perspective of contents are quite low.
- We need to redesign a specialized search engine for it.
- There are few very good attempts:
  - www.wolframalpha.com
  - symbolab.com
  - searchenginewatch.com

Question & Answering Systems

- System that enables the extraction of an answer (or several) to a request (a question) based on a corpus
- An answer or several, possibly a list from one or several documents, an answer of the type Yes/No…,
  - Askjevees.com
  - Easyask.com
  - Answerlogic.com